

Spin-Filtered Recursion and the Emergence of Spacetime from Dimensional Collapse

Eliza Sulea^{1*}

¹Independent Researcher, —, Wilmington, DE, USA.

Corresponding author(s). E-mail(s): esulea@widener.edu;

Abstract

We propose a unified geometric framework in which spacetime, dimensionality, and quantum behavior emerge from a recursive collapse mechanism governed by spin and curvature dynamics. Starting from a six-dimensional scalar field on a smooth Lorentzian manifold, we introduce a spin-filtered recursive operator algebra that induces deterministic dimensional descent. This process selects three-dimensional spherical geometry as the unique spin-stable attractor, with time arising as the recursion index and entropy as the irreversible decay of dimensional degrees of freedom. The cosmological constant is reinterpreted as geometric surface tension at a non-persistent modulation shell, and quantum interference emerges as residual curvature echoes from sub-survivable dimensions. The framework predicts empirically testable features in the cosmic microwave background, including large-angle power suppression, parity asymmetry, and low- ℓ mode alignment—arising naturally from curvature-filtered recursion. Unlike anthropic or inflationary approaches, this model treats dimensionality and temporal flow not as fundamental inputs but as emergent consequences of internal geometric constraints. The universe, in this view, is the stabilized residue of recursive annihilation: a spin-governed sieve through which only curvature-sustaining structure survives.

Keywords: spin-filtered recursion, dimensional collapse, emergent spacetime, curvature dynamics, recursive geometry, parity asymmetry

1 Introduction

In a series of prior works, we have constructed a unified framework in which the observable universe emerges from a recursive geometric process involving higher-dimensional structure. Within this model, curvature, interference, and dimensional collapse together generate the phenomena typically associated with quantum behavior, dark energy, and dimensionality itself [2, 3, 7].

In *Surface Tension from Curved Hypersurfaces as the Origin of the Cosmological Constant*, we aimed to demonstrate that the cosmological constant Λ can be reinterpreted as surface tension arising from the curvature of a 3D hypersurface embedded in a higher-dimensional space. This formulation would eliminate the need for vacuum energy, instead modeling Λ as an emergent equilibrium pressure associated with geometric structure [2, 3].

In *Quantum Behavior as Projected Interference from Higher-Dimensional Scalar Fields*, we introduced the notion of the observable universe experiencing resonant interference between scalar fields in overlapping dimensional spaces. Quantum interference and localization were shown to emerge naturally from the curvature-modulated interaction of a 6D ambient field with a confined 3D substructure [17, 18].

In *Recursive Projection and Dimensional Collapse*, we proposed that spatial dimensionality is not a fundamental input but an emergent outcome of recursive filtering. Using a parity-based recursive rule, we aimed to show that all dimensions except three collapse through successive iterations. Dimension three emerges as a structural attractor, uniquely capable of sustaining curvature without singularity [2, 7].

In this work, we extend that model by introducing *spin* as the internal mechanism governing recursive dimensional filtering, and we define *time* as the iteration count over these spin-governed cycles. We show that this recursive mechanism leads inevitably to dimensional collapse, with only 3D spherical geometry surviving. Importantly, we aim to show that the collapse is not merely topological, but rather curvature-driven, and thus only 3D spheres maintain spin-stable curvature under recursive iteration. Higher-dimensional spin configurations either destabilize or fail to close, resulting in collapse [15, 18].

We further show that the 6D scalar field does not collapse into persistent spatial volume, but into a dynamic curvature modulation shell that surrounds the stabilized 3D geometry. This shell governs recursive behavior but does not appear to be itself geometrically persistent. Alternation effects—phenomena, traditionally interpreted as quantum weirdness, are reinterpreted here as echoes from collapsing dimensions (transient artifacts of ongoing geometric recursion) [18].

In this framework, time is not a primitive parameter. It arises as a discrete index over recursive spin-filtered steps. Entropy is reinterpreted as the decay of dimensional degrees of freedom, and the observable universe is characterized as the fixed-point 3D attractor of a recursive collapse from a higher-dimensional field into a spin-stable 3D hypersurface [4, 14, 17].

The result is a unified formalism in which dimensionality, time, and structure are no longer taken as fundamental, but instead emerge from a process of recursive geometric collapse driven by spin and curvature dynamics.

2 Recursive Spin Filtering and Dimensional Survival

We formalize a variational mechanism through which recursive spin-filtered geometric recursion of a higher-dimensional scalar field induces deterministic dimensional collapse. Three-dimensional geometry emerges as a unique curvature-preserving attractor. In this framework, time, entropy, and geometric stability are not assumed as primitives but arise as consequences of spin-topological constraints encoded within the recursive operator algebra.

2.1 Geometric Action and Variational Collapse Formalism

Let $\mathcal{M}^{(6)}$ be a 6-dimensional smooth Lorentzian manifold with flat metric η_{AB} and a globally trivial tangent bundle. Define a real scalar field $\phi : \mathcal{M}^{(6)} \rightarrow \mathbb{R}$. The standard bulk action is:

$$S_{\text{bulk}}[\phi] = \int_{\mathcal{M}^{(6)}} -\frac{1}{2} \eta^{AB} \partial_A \phi \partial_B \phi d^6 x. \quad (1)$$

We introduce a curvature-functional recursive collapse action:

$$S_{\text{collapse}} = \sum_n \int_{\mathcal{M}^{(d_n)}} \left[-\mathcal{V}_{\text{geom}}(\mathcal{R}^{(d_n)}, \nabla \phi^{(d_n)}) \right] d^{d_n} x, \quad (2)$$

where $\mathcal{R}^{(d)}$ is the Ricci scalar of $\mathcal{M}^{(d)}$, and the collapse potential is defined by:

$$\mathcal{V}_{\text{geom}} := \lambda_n \Theta(\Delta \mathcal{R} - \mathcal{R}_c) \left[\nabla^A \phi^{(d)} \nabla_A \phi^{(d)} \right] \cdot (\Delta \mathcal{R})^2, \quad (3)$$

with $\Delta \mathcal{R} := \|\mathcal{R}^{(d)} - \bar{\mathcal{R}}^{(d-1)}\|$, measuring deviation from curvature continuity. Collapse is triggered when geometric tension exceeds the critical threshold $\mathcal{R}_c$, derivable from Gauss–Bonnet–Chern topological invariants [15]:

$$\mathcal{R}_c = \frac{1}{G_N} \int_{\mathcal{M}^{(d)}} \mathcal{G}^{(d)} dV, \quad (4)$$

where $\mathcal{G}^{(d)}$ is the Gauss–Bonnet density in dimension d .

We take the L^2 norm over $\mathcal{M}^{(d)}$, unless otherwise noted. Although the collapse potential $\mathcal{V}_{\text{geom}}$ is defined variationally, we can describe an effective dynamics via a recursion-gradient flow equation:

$$\frac{d\phi^{(d)}}{dt} = -\frac{\delta S_{\text{collapse}}}{\delta \phi^{(d)}}, \quad (5)$$

where t is the recursion index as defined in Eq. (12). This equation describes dimensional collapse as a dissipative flow in configuration space, wherein unstable spin and curvature modes decay toward the spin-stable attractor at $d = 3$ [17]. The recursion operator $\mathcal{P}_n$ may then be interpreted as a time-discretized descent step within this gradient process.

2.2 Recursive Operator Algebra $\mathcal{P}$

Let $\varphi^{(d)} := \phi^{(d)}$. Define the recursive geometry algebra $\mathcal{P} := \{P_n : \mathcal{H}^{(d_n)} \rightarrow \mathcal{H}^{(d_{n+1})}\}$, where $\mathcal{H}^{(d)}$ is the Hilbert space of square-integrable scalar fields over $\mathcal{M}^{(d)}$. Each P_n is a non-unitary, non-invertible bounded operator with the decomposition:

$$P_n = \Pi_{\text{Spin}}^{(d_n)} \circ \Pi_{\text{Curv}}^{(d_n)} \circ \Pi_{\text{Harm}}^{(d_n)}, \quad (6)$$

where: - $\Pi_{\text{Harm}}^{(d)}$: projects onto hyperspherical harmonics on S^d , - $\Pi_{\text{Curv}}^{(d)}$: filters by curvature eigenmode stability, - $\Pi_{\text{Spin}}^{(d)}$: filters by spin closure under Clifford module representation.

Let the spectral basis be $\{Y_{\ell m}^{(d)}\}$, with:

$$P_n[\phi^{(d)}](x) = \sum_{\ell, m} Y_{\ell m}^{(d)}(x) \cdot \mathcal{F}_{\ell}^{(d)}[\phi^{(d)}], \quad (7)$$

where the filter function satisfies:

$$\mathcal{F}_{\ell}^{(d)}[\phi] = \begin{cases} \phi, & \text{if } \sigma(\mathcal{R}_{\ell}) < \delta_c \text{ and } \rho_{\text{Spin}(d)}(\phi) = \phi, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

2.3 Spin-Stable Dimensionality from Clifford Closure

Let $\text{Cl}(d)$ be the Clifford algebra over $\mathbb{R}^d$, and $\rho : \text{Spin}(d) \rightarrow \text{Aut}(\phi^{(d)})$ a representation acting on field configurations. Spin-stability requires:

$$\rho(R)[\phi] = \phi \quad \forall R \in \text{Spin}(d), \quad (9)$$

with Clifford closure under:

$$[\gamma_{\mu}, \gamma_{\nu}]_+ = 2\eta_{\mu\nu} \mathbb{I}, \quad \mu, \nu \in \{1, \dots, d\}. \quad (10)$$

The only dimension for which curvature-compatible spin-preserving Clifford action admits recursive closure is:

$$d = 3[7]. \quad (11)$$

2.4 Dimensional Collapse as Recursion-Indexed Flow

Define dimensional evolution as a recursion function:

$$d_{n+1} = f(d_n) := \begin{cases} d_n - 2, & \text{if } d_n \equiv 0 \pmod{2}, d_n \neq 2, \\ d_n - 1, & \text{if } d_n \equiv 1 \pmod{2}, d_n \neq 3, \\ d_n, & \text{if } d_n = 3. \end{cases} \quad (12)$$

This generates a monotonic descent terminating in a 3D fixed point:

$$\exists N \in \mathbb{N} \text{ such that } \forall n \geq N, d_n = 3.$$

2.5 Time and Entropy from Recursive Topology

Define emergent time as the recursion index:

$$t := n. \quad (13)$$

Let $\mathcal{V}^{(d)}$ be the volume of $\mathcal{M}^{(d)}$, and define entropy flow as:

$$\frac{dS}{dt} := \log \left(\frac{\mathcal{V}^{(d_n)}}{\mathcal{V}^{(d_{n+1})}} \right) > 0. \quad (14)$$

This captures the effective information loss through dimensional reduction, establishing a geometric arrow of time.

2.6 Modulation Shell as Cohomological Tension Layer

Define the modulation shell as the recursive boundary residue:

$$\Sigma^{(d)} := \partial \left(\mathcal{M}^{(d)} \setminus \mathcal{M}^{(d-1)} \right), \quad (15)$$

and define a tension scalar $\psi : \Sigma^{(d)} \rightarrow \mathbb{R}$ via:

$$\psi(x) := \nabla \phi^{(d)} \cdot \nabla \phi^{(d-1)} \Big|_{\Sigma^{(d)}}. \quad (16)$$

We interpret the modulation shell $\Sigma^{(d)}$ as a cohomologically active boundary layer encoding residual curvature memory. It satisfies a discontinuous recursion equation:

$$\nabla^2 \psi = \delta^{(d)}(\Sigma^{(d)}) \cdot \kappa(x), \quad (17)$$

where $\delta^{(d)}$ is a shell-supported Dirac distribution and $\kappa(x)$ encodes local curvature tension. This suggests that $\Sigma^{(d)}$ behaves like a dynamical brane or geometric memory surface. Future work may explore whether ψ admits a spin-cohomology class under $H_{\text{Spin}}^1(\Sigma^{(d)}; \mathbb{R})$, potentially linking this structure to persistent topological modes.

Since the scalar field satisfies a discontinuous PDE across the shell, it may admit a nontrivial de Rham cohomology class:

$$[\psi] \in H^1(\Sigma^{(d)}; \mathbb{R}),$$

encoding curvature memory from recursive dimensional descent.

2.7 Terminal Recursion and the Observed Universe

We define the observed universe as:

$$\mathcal{U}_{\text{obs}} := \lim_{n \rightarrow \infty} P_n[\phi^{(6)}] \Big|_{d=3} \cup \bigcup_{d_n \in \{1,2\}} \lim_{n \rightarrow \infty} \psi^{(d_n)}, \quad (18)$$

where residual modes $\psi^{(1,2)}$ act as curvature-encoded harmonic perturbations, generating interference-like features within the stabilized 3D recursion manifold.

This formulation does not posit a holographic duality between bulk and boundary. Instead, the observed universe arises as a recursive geometric fixed point, stabilized by spin-filtered dimensional collapse. The residual fields are not dual images. They are persistent curvature residues from sub-survivable dimensions.

3 Observational Signatures of Recursive Dimensional Collapse in the CMB

The recursive collapse mechanism proposed in this work predicts a potential spectrum of empirical consequences for the cosmic microwave background (CMB). As spin-filtered descent proceeds from $d = 6$ to $d = 3$, intermediate recursion layers and residual harmonic shells induce non-Gaussian, large-angle anisotropies in the post-recombination radiation field. This section outlines specific observable quantities that may perhaps validate or constrain the dimensional recursion model.

3.1 Large-Scale Power Suppression from Early Curvature Damping

The initial collapse steps from $d = 6 \rightarrow 5 \rightarrow 4$ produce high-curvature regimes whose unstable spin modes are filtered by the recursive operator P_n . This filtration reduces spectral support in low- ℓ curvature harmonics, resulting in suppressed Sachs–Wolfe temperature variance at large angular scales:

$$C_\ell = \langle |a_{\ell m}|^2 \rangle \quad \text{with } C_2, C_3 < \Lambda\text{CDM}_{\text{mean}} - 2\sigma. \quad (19)$$

This naturally may explain the empirically observed low quadrupole and octupole moments reported in *Planck 2018* results [12, 19].

We define the harmonic suppression function induced by recursive dimensional collapse as:

$$S_\ell^{\text{rec}} = \exp(-\alpha \cdot \ell(\ell+1)/\ell_{\text{cut}}^2), \quad (20)$$

where α is a curvature damping parameter and $\ell_{\text{cut}} \sim 30$ defines the suppression scale from early high-curvature modes. The resulting CMB temperature spectrum is modified as:

$$C_\ell^{\text{rec}} = C_\ell^{\Lambda\text{CDM}} \cdot S_\ell^{\text{rec}}. \quad (21)$$

This again, may naturally reproduces the observed suppression at $\ell = 2, 3$ without fine-tuning.

To model the parity asymmetry, we define the filtered spectrum decomposition:

$$C_\ell^{\text{even}} = C_\ell \cdot \delta_{\ell \bmod 2, 0}, \quad C_\ell^{\text{odd}} = C_\ell \cdot \delta_{\ell \bmod 2, 1}, \quad (22)$$

and rewrite the asymmetry index P_ℓ in terms of filtered residues. Collapse-induced parity violation predicts $P_\ell < 0$ with maximal asymmetry near $\ell \sim 10$.

3.2 Parity Violation and Hemisphere Asymmetry from Spin-Broken Collapse

The dimensional reduction rule breaks discrete spatial symmetries by construction. Parity is explicitly violated in two respects:

1. The collapse rule is not invariant under spatial inversion $\vec{x} \rightarrow -\vec{x}$, due to spin-closure constraints favoring one orientation of recursion.
2. The modulation shell $\Sigma^{(d)}$ produces anisotropic boundary conditions during collapse, which inject parity-odd signatures into the residual curvature field.

These effects generate a measurable hemispherical asymmetry in both temperature and polarization fields, which can be quantified via the parity asymmetry parameter $\mathcal{P}_\ell$:

$$\mathcal{P}_\ell = \frac{\sum_{\ell=2}^{\ell_{\max}} C_\ell^{\text{even}} - C_\ell^{\text{odd}}}{\sum_{\ell=2}^{\ell_{\max}} C_\ell^{\text{even}} + C_\ell^{\text{odd}}}, \quad \text{with } \mathcal{P}_\ell < 0 \text{ expected.} \quad (23)$$

This seems to be consistent with findings in [8, 22], which show significant parity asymmetry at $\ell < 30$.

3.3 Axis of Evil from Anisotropic Modulation Shell Collapse

The emergent 3D geometry in this model would hypothetically inherit topological scars from directional curvature gradients during recursion. This anisotropy manifests as an apparent alignment of the low- ℓ multipoles, infamously dubbed the “axis of evil.” In our model, this results from asymmetric collapse dynamics of the $d = 5 \rightarrow 4$ modulation shell:

$$\hat{n}_{\max} = \arg \max_{\hat{n}} \sum_{\ell=2}^{\ell=3} |a_{\ell m}(\hat{n})|^2. \quad (24)$$

If the collapse has a preferred curvature flow direction, it imprints a statistically anomalous axis across the quadrupole and octupole moments [6, 19].

3.4 Residual Harmonic Echoes from Lower-Dimensional Collapse States

The final recursion limit leaves behind residual harmonic structures from the $d = 1, 2$ boundary states:

$$\text{Reality} = \lim_{n \rightarrow \infty} P_n[\phi^{(6)}] \Big|_{d=3} \cup \lim_{n \rightarrow \infty} \{\text{Residuals from } d_n \in \{1, 2\}\}. \quad (25)$$

These residues generate long-wavelength interference-like signatures, potentially detectable via the angular correlation function $C(\theta)$:

$$C(\theta) = \sum_{\ell} \frac{2\ell+1}{4\pi} C_\ell P_\ell(\cos \theta), \quad \text{with } C(\theta > 60^\circ) < \Lambda\text{CDM}_{\text{pred}}. \quad (26)$$

This matches the observed suppression of large-angle correlations (the “ $S_{1/2}$ anomaly”) noted by both *WMAP* and *Planck* [12, 19].

These observations are circumstantial at this time and further experiments targeting this theory would need to be explored.

3.5 Predicted Signatures and Empirical Tests

The framework outlined here appears to suggest several observational signatures that could, in principle, offer a means of empirical testing. One such potential indicator is a persistent suppression of power in the CMB temperature power spectrum at low multipoles, specifically for $\ell \leq 30$. While this feature has been noted in previous observations [12, 19], its continued presence across future datasets would strengthen the case for a non-standard explanation.

In addition, a systematically negative parity asymmetry index $\mathcal{P}_\ell$ may be observable in full-sky analyses. If such asymmetry persists and is not attributable to instrumental or foreground contamination, it might reflect a deeper geometric origin [19].

There is also some indication that the quadrupole and octupole modes exhibit anomalous alignment across successive datasets [6, 19]. Although the statistical significance of this alignment remains debated, it could potentially serve as a nontrivial signature of recursive structure if it proves robust under improved measurements.

Furthermore, a notable reduction in the two-point angular correlation function at large angular scales (for $\theta > 60^\circ$) may continue to challenge the expectations of Gaussian random field models [12]. This feature, if confirmed with higher-precision data, would pose a significant constraint on standard cosmological assumptions.

Finally, there may be weak, scale-dependent residual correlations in the polarization channels, particularly in the EE and TE spectra. While evidence for this is currently limited, future studies could explore whether such correlations arise naturally within spin-filtered recursive transfer functions. Targeted simulations and comparisons with data from *Planck*, *LiteBIRD*, and *CMB-S4* would be essential for testing this possibility [12].

Empirical Falsifiability and Future Constraints

Despite its speculative nature, the model outlined here leads to a set of testable predictions that distinguish it from both the standard Λ CDM framework and anthropic multiverse approaches [23]. If future observations indicate a full recovery of low- ℓ CMB power consistent with Gaussian Λ CDM predictions, particularly in missions such as *LiteBIRD* and *CMB-S4*, then the viability of this model would be substantially diminished [12].

Similarly, the absence of a significant large-angle parity asymmetry (if $\mathcal{P}_\ell \rightarrow 0$) in upcoming high-fidelity polarization data would challenge one of the framework's central expectations [19]. The lack of statistically meaningful alignment between the quadrupole and octupole axes in future multi-year surveys would also weigh against the hypothesis of recursive geometric ordering [6, 19].

Finally, if no suppression of long-range angular correlations is observed beyond $\theta > 60^\circ$, and the angular correlation function aligns fully with predictions from standard cosmology, this would constitute strong evidence against the model [12].

Unlike inflationary or multiverse-based explanations that often invoke inaccessible or highly contingent initial conditions [23], this approach attempts to offer a more direct geometric account of observed anomalies. Whether this effort ultimately succeeds remains to be seen, but it does lend itself to near-future observational constraints in a way that few alternative proposals currently do.

4 Discussion

The model presented in this work advances a novel synthesis of dimensional collapse, spin dynamics, and emergent spacetime. At its core lies a departure from conventional assumptions about dimensionality, treating it not as a fixed background structure but as an unstable, recursive phenomenon shaped by spin-mediated filtering. This perspective enables a reinterpretation of space, time, and entropy as emergent outcomes of recursive geometric descent rather than primitive inputs to physical theory.

Whereas many contemporary approaches to quantum gravity, such as causal dynamical triangulations [2, 10], asymptotic safety [16], and loop quantum gravity [18], treat dimensional reduction as a coarse-grained limit or statistical artifact, the present framework derives it as a deterministic consequence of spin-instability in higher dimensions. The recursive collapse rule defined here is not a renormalization effect, nor an emergent average, but a direct geometric mechanism governed by curvature continuity and spin closure. This distinguishes the model sharply from prior formulations, offering an intrinsic pathway for the emergence of a 3-dimensional universe.

In particular, the identification of the 2-sphere (embedded in $\mathbb{R}^3$) as the unique curvature-stable manifold under recursive spin filtering provides a geometric rationale for the persistence of exactly three spatial dimensions. The spin stability criterion introduced here, requiring smooth, axis-preserving, curvature-closing rotation, is not satisfied in higher-dimensional settings due to torsional divergence or failure of closure. This result parallels, but ultimately extends, dimensional selection mechanisms proposed in early string gas cosmology [5] and evolving dimension frameworks [1], by replacing probabilistic arguments with a structural collapse process.

The model’s treatment of time as the index over recursion cycles also offers a conceptually clean resolution to the arrow of time. Rather than invoking asymmetry through initial conditions or external parameters, time in this formulation emerges from the irreversible sequence of dimensional descent. As spin recursion eliminates higher-dimensional degrees of freedom, entropy increases monotonically by geometric necessity. This bears structural resemblance to time-asymmetry in Einstein–Cartan cosmology [13], where spin and torsion define irreversibility, though here the mechanism arises from recursive collapse rather than bounce dynamics. Microphysical support for this view can be found in recent experimental studies of thermodynamic irreversibility in isolated spin systems [20].

Although the language of holography provides useful metaphor, this model departs from standard dualities. While projecting a 6D scalar field onto a 3D surface resembles the structure of AdS/CFT [11], here the recursion is not a duality. Rather, it is an internal, dynamic collapse within geometry itself. The higher-dimensional “modulation shell” acts not as a boundary encoding, but as a transient curvature regulator that

governs recursive stability. In this way, the model suggests a deeper unification between holographic logic and spin-driven dimensional collapse, not via correspondence but through structural recursion. It may perhaps offer a bridge between geometric ontology and quantum informational architecture [4].

Furthermore, this framework offers a geometric reinterpretation of quantum interference. Alternating recursion amplitudes from collapsing dimensions generate non-persistent but structured transients as residual patterns that mirror quantum indeterminacy. These interference-like effects may arise, not from superposition, but from phase modulation encoded in recursive curvature decay. This view seems to resonate with emergent spacetime models in which entanglement structure defines geometry [24], but reframes that connection as a consequence of dimensional recursion, not fundamental dualism.

Despite its conceptual coherence, several limitations remain. The recursive operator $\mathcal{P}_n$ is currently defined algebraically, not yet embedded in a full dynamical field-theoretic framework. A Lagrangian or Hamiltonian formulation capable of encoding the curvature-driven recursion dynamics would be an important next step. Similarly, the curvature threshold κ_c is introduced heuristically. A derivation from first principles—perhaps via topological invariants or gravitational entropy would strengthen the model’s foundations. The modulation shell remains an open object. While defined as the residue of collapse, its physical role (as a field, membrane, or geometrically regulated transition layer) awaits further mathematical elaboration, possibly through noncommutative geometry or quantum topology.

Preliminary analysis suggests that the modulation shell $\Sigma^{(d)}$ may admit a formal description via spin cobordism and index theory. Its discontinuous behavior and memory of collapsed curvature could allow classification through the Atiyah–Singer index theorem or residue trace formulations. Interpreting $\Sigma^{(d)}$ as a spin^c boundary layer may link collapse topology to residual topological field content, and potentially connect with spectral triple frameworks in holographic noncommutative geometry.

Although the model already offers testable structural predictions, such as the uniqueness of 3D as a spin-stable attractor and the emergence of time from recursive descent, it does not yet produce specific observational signatures from first principles. Nonetheless, this framework may perhaps open rich territory for phenomenological exploration such as cosmic isotropy violations, parity asymmetries in galactic spin distributions [21], and fractal geometry in CMB anomalies. They may all serve as empirical footholds for recursive dimensional collapse.

In sum, this work proposes a foundational shift in our understanding of space, time, and structure, not as ontologically primary, but as emergent from recursive spin-governed descent. By synthesizing insights from geometry, quantum theory, holography, and cosmology, it aims to recast dimensional collapse as the generative act of physical law. The resulting universe thus, may not be a passive solution to initial conditions, but the stabilized residue of a recursive collapse process, or rather, a geometric filtration through which only structure capable of spin-stable curvature remains.

5 Conclusion

This work aims to propose a unified geometric framework in which spacetime, time, and quantum behavior arise from a process of recursive dimensional collapse governed by spin-filtration. While departing from traditional models that treat dimensionality and temporal flow as fundamental, we have shown that these features can be derived from internal consistency conditions. Specifically, this work focuses on curvature continuity and spin closure, within a collapsing higher-dimensional scalar field. Dimensional survival thus, in this model, is not arbitrary but the result of a strict selection mechanism, with 3D curvature emerging as a unique attractor capable of sustaining spin-stable recursion.

Time, in this framework, is no longer a background parameter but the indexing of recursive descent itself, while entropy corresponds to the irreversible loss of dimensional degrees of freedom. The cosmological constant, rather than arising from vacuum energy, is reinterpreted as geometric surface tension induced by residual curvature at the modulation shell. Observable quantum effects, including interference and nonlocality, are reframed as geometric transients; the echoes of recursive instability from higher dimensions that did not survive collapse.

Our analysis further connects this collapse engine to cosmological observables, offering falsifiable predictions for CMB anisotropies, parity asymmetries, and angular correlation anomalies. These features emerge naturally from the recursive filtering process, without the need for anthropic assumptions, initial condition fine-tuning, or multiverse inflationary speculation.

Ultimately, the framework reimagines the universe not as a static structure or preordained manifold, but as the stabilized remnant of a recursive descent (a spin-governed sieve through which only the geometrically coherent can persist). This collapse, once completed, may define what we call reality. In this sense, reality would be a dynamically selected, curvature-sustaining structure that embodies the endpoint of recursive annihilation. Thus, in this speculative framework, it is not spacetime that gives rise to structure, but structure filtered by spin—that gives rise to spacetime.

Declarations

- **Funding:** This work received no external funding.
- **Conflict of interest/Competing interests:** The author declares no competing financial or non-financial interests.
- **Ethics approval and consent to participate:** Not applicable.
- **Consent for publication:** Not applicable.
- **Data availability:** All data used are from publicly available sources, as cited in the references. No new datasets were generated.
- **Materials availability:** Not applicable.
- **Code availability:** No code was developed or used in this study.
- **Author contribution:** The author is solely responsible for the conception, derivation, analysis, writing, and editing of this manuscript.
- **Use of AI tools:** AI (ChatGPT-4o, OpenAI) was used under the author’s direction to assist with mathematical derivations, symbolic computation, and conceptual

clarification of physics topics. The author independently conceived the core ideas, selected and devised the appropriate models, interpreted all results, and wrote the paper. AI was used as an educational and computational aid; it did not generate the scientific content, arguments, or conclusions without author oversight.

Editorial Policies for:

Springer journals and proceedings: <https://www.springer.com/gp/editorial-policies>

Nature Portfolio journals: <https://www.nature.com/nature-research/editorial-policies>

Scientific Reports: <https://www.nature.com/srep/journal-policies/editorial-policies>

BMC journals: <https://www.biomedcentral.com/getpublished/editorial-policies>

Appendix A Speculations

Recursive Dimensional Collapse and the Landau-Level Analogy

The recursive dimensional collapse model described in this paper exhibits structural behavior that closely mirrors the known quantum dynamics of electrons in magnetic fields, particularly in the context of Landau quantization. In a strong magnetic field, electrons are confined to discrete, spin-dependent energy levels (Landau levels) whose occupation is governed by spin alignment and field intensity. Higher energy states may decay or collapse due to instability, while the lowest-energy ground state remains stably occupied [9].

This behavior is isomorphic to the dimensional recursion engine described herein. Each dimension in the recursive system is filtered through a spin-compatible collapse rule, where higher-dimensional configurations tend toward instability. Intermediate states alternate recursively, and only a single dimensional layer, namely, three-dimensional curvature, exhibits long-term geometric stability.

In the Landau system [9], electrons collapse through quantized energy levels until a stable spin-aligned state is reached. In the recursive framework, dimensions collapse through quantized, parity-sensitive recursion steps until a spin-compatible curvature shell (dimension 3) stabilizes. This analogy is not metaphorical but structural as both systems seem emerge from recursive descent through quantized states governed by spin constraints.

This structural equivalence may suggest that the recursive collapse engine is not merely a dimensional artifact but may represent a universal stabilizing principle. Electrons in naturally occurring field structures (across stars, magnetospheres, and vacuum fluctuations) behave analogously, implying that spin-governed collapse may be a ubiquitous mechanism in physical systems. By extension, if such collapse engines can spontaneously stabilize under the right geometric constraints, then the emergence

of spin-compatible “reality structures” may not be singular, but simply rare, implying a multiversal landscape populated not by infinite randomness but by a set of constrained, recursive survivors [23].

Directionality, Spin Symmetry, and the Antimatter Window

Within the collapse model, spin functions not merely as a particle property but as a directional recursive operator. It is a mechanism for selecting the orientation of dimensional recursion. Once spin is established in a coherent recursive direction, a time vector is induced and geometric stability emerges. However, the system does not inherently require a specific spin direction; it merely requires consistency.

This observation may suggest that during dimensional recursion, either spin orientation could be selected (clockwise or counterclockwise) relative to the recursion index, and as long as the orientation is preserved, stable structure can emerge. Consequently, matter and antimatter may represent mirrored collapse trajectories, each stabilized by its own recursive spin vector. In this framing, antimatter is not the inverse of matter in content, but in recursion orientation.

This raises the possibility that alternate universes (stabilized collapse events with opposite spin parity as per this model) may exist as phase-displaced counterparts to our own. They would not be observable through direct coupling because they exist on a reversed recursion path. The present model offers a structural foundation for this interpretation, proposing that the observed matter-antimatter asymmetry in our universe is a consequence of collapse directionality, not intrinsic imbalance. Antimatter, therefore, becomes a conceptual window into the opposite side of the spin recursion engine.

On the Structural Consequences of Spin-Based Recursion

The recursive dimensional framework presented herein allows us to reinterpret several classical physical principles in light of spin-governed structural selection. Foremost among them is the arrow of time, which emerges naturally from the unidirectional indexing of spin-based recursion. In this model, once a spin orientation is selected, recursion proceeds coherently, and temporal asymmetry arises as a structural consequence.

This coherence is both necessary and sufficient for geometric stability. Incoherent or fluctuating spin orientation across recursive steps would fail to sustain curvature continuity and collapse prematurely into lower-dimensional residues. Thus, spin is not merely directional, but rather constitutive. Time, matter, and dimensional stability co-emerge from the persistence of spin alignment.

More significantly, this framework suggests that our universe’s apparent direction of time and matter dominance is not absolute, but rather a selected and preserved outcome of recursive coherence. Other spin orientations could (and may have) stabilized elsewhere under identical constraints but with reversed parity. This reframes our

universe not as uniquely tuned, but as one of many stable outcomes in a constraint-driven spin landscape. The “faucet” of existence may emit countless spin attempts, but only those that stabilize through recursive coherence become real.

References

- [1] Afshordi, N., Stojkovic, D.: Emergent Spacetime in Stochastically Evolving Dimensions. *Physics Letters B* **739**, 117–124 (2014). Reprinted as arXiv:1405.3297 [hep-th]. <https://doi.org/10.1016/j.physletb.2014.10.048>, <https://doi.org/10.48550/arXiv.1405.3297>
- [2] Ambjørn, J., Jurkiewicz, J., Loll, R.: Reconstructing the Universe. *Physical Review D* **72**, 064014 (2005). Reprinted as arXiv:hep-th/0505154. <https://doi.org/10.1103/PhysRevD.72.064014>, <https://doi.org/10.48550/arXiv.hep-th/0505154>
- [3] Arkani-Hamed, N., Dimopoulos, S., Dvali, G.: The hierarchy problem and new dimensions at a millimeter. *Physics Letters B* **429**(3–4), 263–272 (1998). [https://doi.org/10.1016/S0370-2693\(98\)00466-3](https://doi.org/10.1016/S0370-2693(98)00466-3)
- [4] Bousso, R.: The holographic principle. *Reviews of Modern Physics* **74**, 825–874 (2002). Reprinted as arXiv:hep-th/0203101. <https://doi.org/10.1103/RevModPhys.74.825>, <https://doi.org/10.48550/arXiv.hep-th/0203101>
- [5] Brandenberger, R.H., Vafa, C.: Superstrings in the Early Universe. *Nuclear Physics B* **316**, 391–410 (1989). [https://doi.org/10.1016/0550-3213\(89\)90037-0](https://doi.org/10.1016/0550-3213(89)90037-0)
- [6] Eriksen, H.K., Hansen, F.K., Banday, A.J., Górski, K.M., Lilje, P.B.: Asymmetries in the Cosmic Microwave Background Anisotropy Field. *The Astrophysical Journal* **605**(1), 14–20 (2004). <https://doi.org/10.1086/382267>
- [7] Furey, C.: Three generations, two unbroken gauge symmetries, and one eight-dimensional algebra. *Physics Letters B* **785**, 84–89 (2018). <https://doi.org/10.1016/j.physletb.2018.08.032>
- [8] Kim, J., Naselsky, P.: Anomalous parity asymmetry of the Wilkinson Microwave Anisotropy Probe power spectrum data at low multipoles. *Astrophysical Journal Letters* **714**, L265–L267 (2010). Reprinted as arXiv:1001.4613 [astro-ph.CO]. <https://doi.org/10.1088/2041-8205/714/2/L265>, <https://doi.org/10.48550/arXiv.1001.4613>
- [9] Landau, L.D., Lifshitz, L.M.: *Quantum Mechanics: Non-Relativistic Theory*, 3rd edn. Butterworth-Heinemann, Oxford (1981). ISBN: 978-0750635394
- [10] Loll, R.: Quantum Gravity from Causal Dynamical Triangulations: A Review. *Classical and Quantum Gravity* **37**(1), 013002 (2020). Reprinted

- as arXiv:1905.08669 [hep-th]. <https://doi.org/10.1088/1361-6382/ab57c7>,
<https://doi.org/10.48550/arXiv.1905.08669>
- [11] Maldacena, J.M.: The Large N Limit of Superconformal Field Theories and Supergravity. *Advances in Theoretical and Mathematical Physics* **2**, 231–252 (1998). Reprinted as arXiv:hep-th/9711200. <https://doi.org/10.1023/A:1026654312961>,
<https://doi.org/10.48550/arXiv.hep-th/9711200>
 - [12] Planck Collaboration, Aghanim, N., et al.: Planck 2018 results. VI. Cosmological parameters. *Astron. Astrophys.* **641**, A6 (2020). <https://doi.org/10.1051/0004-6361/201833910>
 - [13] Popławski, N.J.: Cosmology with torsion: An alternative to cosmic inflation. *Physics Letters B* **694**, 181–185 (2010); Erratum: *ibid.* **701**, 672 (2011). Reprinted as arXiv:1007.0587 [astro-ph.CO]. <https://doi.org/10.1016/j.physletb.2010.09.056>,
<https://doi.org/10.48550/arXiv.1007.0587>
 - [14] Prigogine, I.: *From Being to Becoming: Time and Complexity in the Physical Sciences*. W.H. Freeman & Co., San Francisco (1981). ISBN: 978-0716711087
 - [15] Reed, M., Simon, B.: *Functional Analysis*. Methods of Modern Mathematical Physics, vol. 1, Revised edn. Academic Press, New York (1980). ISBN: 978-0125850506
 - [16] Reuter, M., Saueressig, F.: Quantum Einstein Gravity. *New Journal of Physics* **14**, 055022 (2012). Reprinted as arXiv:1202.2274 [hep-th]. <https://doi.org/10.1088/1367-2630/14/5/055022>,
<https://doi.org/10.48550/arXiv.1202.2274>
 - [17] Connes, A., Rovelli, C.: Von Neumann Algebra Automorphisms and Time–Thermodynamics Relation in General Covariant Quantum Theories. *Classical and Quantum Gravity* **11**(12), 2899–2918 (1994). Reprinted as arXiv:gr-qc/9406019. <https://doi.org/10.1088/0264-9381/11/12/007>,
<https://doi.org/10.48550/arXiv.gr-qc/9406019>
 - [18] Rovelli, C.: *Quantum Gravity*. Cambridge Monographs on Mathematical Physics, 1st edn. Cambridge University Press, Cambridge (2007). ISBN: 978-0521715966
 - [19] Schwarz, D.J., Starkman, G.D., Huterer, D., Copi, C.J.: Is the low- ℓ microwave background cosmic? *Physical Review Letters* **93**, 221301 (2004). Reprinted as arXiv:astro-ph/0403353. <https://doi.org/10.1103/PhysRevLett.93.221301>,
<https://doi.org/10.48550/arXiv.astro-ph/0403353>
 - [20] Micadei, K., Peterson, J.P.S., Souza, A.M., Sarthour, R.S., Oliveira, I.S., Landi, G.T., Batalhão, T.B., Serra, R.M., Lutz, E.: Reversing the thermodynamic arrow of time using quantum correlations. *arXiv:1711.03323* [quant-ph] (2017).

<https://doi.org/10.48550/arXiv.1711.03323>

- [21] Shamir, L.: New evidence and analysis of cosmological-scale asymmetry in galaxy spin directions. *Journal of Astrophysics and Astronomy*, accepted (2022). Reprinted as arXiv:2201.03757 [astro-ph.CO]. <https://doi.org/10.1007/s12036-022-09809-8>, <https://doi.org/10.48550/arXiv.2201.03757>
- [22] Shamir, L.: Large-scale asymmetry in the distribution of galaxy spin directions – analysis and reproduction. *Symmetry* **15**(9), 1704 (2023). Reprinted as arXiv:2309.03418 [astro-ph.CO]. <https://doi.org/10.3390/sym15091704>, <https://doi.org/10.48550/arXiv.2309.03418>
- [23] Tegmark, M.: Parallel Universes. In: Barrow, J.D., Davies, P.C.W., Harper, C.L. (eds.) *Science and Ultimate Reality: From Quantum to Cosmos*, pp. 459–491. Cambridge University Press, Cambridge (2003). Reprinted as arXiv:astro-ph/0302131. <https://doi.org/10.48550/arXiv.astro-ph/0302131>
- [24] Van Raamsdonk, M.: Building up spacetime with quantum entanglement. *General Relativity and Gravitation* **42**, 2323–2329 (2010); also in *International Journal of Modern Physics D* **19**, 2429–2435 (2010). Reprinted as arXiv:1005.3035 [hep-th]. <https://doi.org/10.1007/s10714-010-1034-0>, <https://doi.org/10.1142/S0218271810018529>, <https://doi.org/10.48550/arXiv.1005.3035>